The Pentagon, Machine-Checked: Abel's five-term relation for the dilogarithm in Lean 4

Special functions · the dilogarithm

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Statements

Theorem 1 — Dilog.rogersL_fiveterm 3

At a glance

What the paper contributes

- A sorry-free Lean 4 proof of Abel's five-term relation for the Rogers L-function, $L(x)+L(y)=L(xy)+L\left(\frac{x(1-y)}{1-xy}\right)+L\left(\frac{y(1-x)}{1-xy}\right)$ for $x, y \in (0,1)$ (Theorem 1).
- The route is the classical one: the five-term difference has zero derivative in x (nine logarithms collapse into a basis of five), so it is constant, and its value is read off as $x \to 0^+$, where $\log t \cdot \log(1-t) \to 0$ by a squeeze.
- Three reusable blocks on the dilogarithm of the companion paper: the derivative of L on $(0,1)$, $L(t) \to 0$ as $t \to 0^+$, and continuity of L on $(0,1)$.
- The mathematics is classical (Spence, Abel, Rogers); the paper claims the formalization, after a documented search of proof-assistant libraries.

🚀 **Start here: Theorem 1**: Abel's five-term relation for the Rogers L-function on (0,1)².

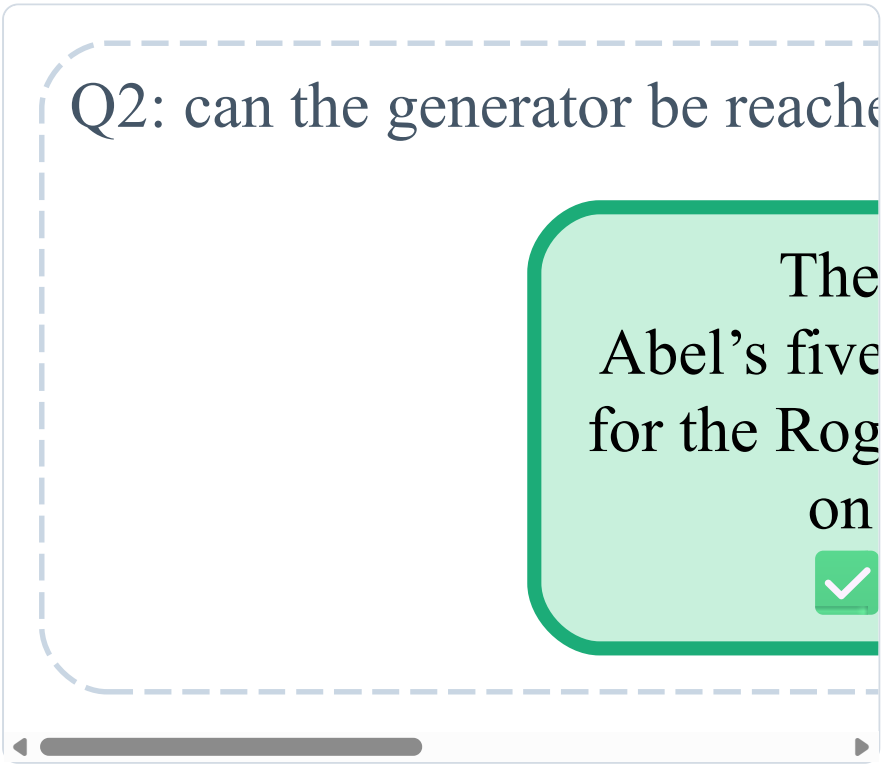
Main results:

- **Theorem 1**: Abel's five-term relation for the Rogers L-function on (0,1)²

1 results | ✅ 1 formalized in Lean | ◐ 0 with a part in Lean | 💻 0 checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



1 Q2: can the generator be reached by the same three-move recipe?

Theorem 1. Lean (`Dilog.rogersL_fiveterm`)✓ p. 3

> *For all* $x, y \in (0,1)$,
>
> $$\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\Big(\tfrac{x(1-y)}{1-xy}\Big) + \mathrm{L}\Big(\tfrac{y(1-x)}{1-xy}\Big).$$

✅ Checked when this page was built: FiveTerm.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — rogersL_fiveterm, fiveterm_hasDerivAt_zero, hasDerivAt_rogersL, tendsto_rogersL_nhdsGT_zero, continuousAt_rogersL: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean FiveTerm.lean` (SHA-256 of the compiled file: `ce29fa490a366ddb…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Spence, Abel, Rogers); the paper claims the formalization: «to the best of my knowledge the first machine-checked five-term relation, on the first machine-checked dilogarithm, in any proof assistant» (§6), from a documented search of June 2026.

↳ Not used by another result of the paper.

Proof

> Formalized in Lean (see the linked declarations).

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